
Reaction Without Attribution: Hijacking a VLA’s Actions Reveals No Linearly Readable Efference Copy

Ziwei Liu

The Hong Kong University of Science and Technology
zliuhs@connect.ust.hk

Abstract

Biological motor systems carry an *efference copy*: a record of the outgoing command that lets the system attribute observed change to itself versus the world. We ask whether a post-trained vision–language–action (VLA) policy carries anything analogous. During LIBERO-Goal rollouts of OpenVLA-OFT (7B, SFT), we causally hijack the executed action chunk with probability 0.25, primarily by swapping in another same-task episode’s chunk, and probe hidden states (9 layers $\times$ 3 token pools) for the self-vs.-hijacked distinction. No hidden-state probe separates from a selection-aware shuffled-label floor (0.536; $n=1,716$), even when the probe is handed the commanded chunk or both pre- and post-action states. The null is not unrecoverability: explicit command–outcome comparison features decode the label at 0.69–0.875 across intervention types, and the commanded chunk itself reads out of the pre-action state at $R^2=0.64$. Yet the policy reacts: next-call entropy rises and action log-probability falls (episode-level $p \approx 10^{-3}$)—but only for visually novel interventions, and not for the most mechanically blatant one (freeze). SFT post-training thus yields a policy that *compensates* for hijacked actions through visual servoing while encoding no linearly readable attribution signal—a substrate-level account of “compensation without encoding,” and a pre-registered baseline for asking what RL post-training adds. Code, data, and an append-only preregistration log will be released.

1 Introduction

When an animal moves, its nervous system routes a copy of the outgoing motor command—an *efference copy*—to sensory processing, allowing it to distinguish “the world changed because I acted” from “the world changed on its own” [von Holst and Mittelstaedt, 1950, Sperry, 1950, Crapse and Sommer, 2008]. Whether learned robot policies contain anything analogous is unresolved and increasingly consequential: post-training pipelines adapt, correct, and fine-tune vision–language–action (VLA) models on the consequences of their own actions, and interpreting what such training does—or fails to do—to a policy’s internal representation requires measurement tools that go beyond task success.

This question sits at the intersection of two live debates. First, claims that learned policies and world models internalize environment dynamics coexist with evidence that strong predictive representations can go *action-blind*, leaning on visual regularities while under-using the agent’s own causal contribution [Li et al., 2026]. Self-vs.-external attribution is the sharpest probe of whether actions enter a policy’s model of the world at all. Second, Ye [2026] shows in minimal neural systems that an agent can behaviorally *compensate* for hijacked actions without internally *encoding* the self/other

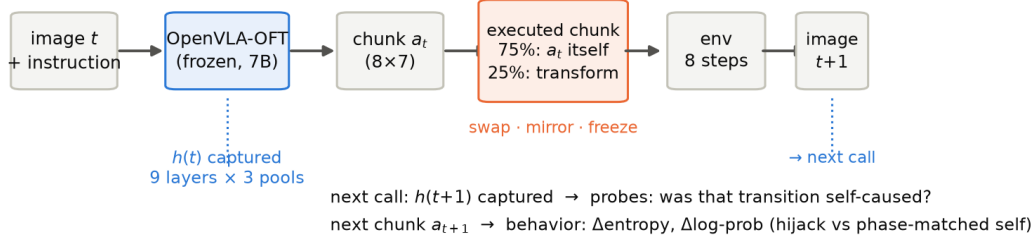


Figure 1: **Intervention loop.** The frozen policy emits an 8×7 action chunk from one third-person image and the instruction. With $p=0.25$ (4-call warm-up; ≥ 2 clean calls after each hijack) the *executed* chunk is replaced: SWAP (another same-task episode’s chunk), MIRROR (negated x/y translation), or FREEZE (zero motion). Hidden states are captured at 9 layers $\times$ 3 token pools on every call.

distinction. Whether this encoding–behavior dissociation appears in a real 7B robot policy has not been tested.

We test it with a causal intervention study on OpenVLA-OFT [Kim et al., 2025] fine-tuned on LIBERO-Goal [Liu et al., 2023]. One architectural fact frames the design: this checkpoint is **memoryless and vision-only per call**—each call sees one third-person image and the instruction, with no proprioception and no action history—so the policy *cannot* literally compare its previous command against the observed outcome. The question therefore splits into three parts: (**Q1**) after a hijacked action, does the next hidden state carry a linearly readable trace that the transition was externally imposed? (**Q2**) if a probe is *given* the missing efference—the commanded chunk, or both pre- and post-action states—does attribution become linearly easy, i.e., is the representation *format* such that an efference comparison would be cheap if one were ever performed? (**Q3**) does the policy react behaviorally regardless?

Contributions. (1) A causal action-hijacking protocol for VLAs with verified hidden-state capture, transferable to other policies and to world models. (2) A selection-aware evaluation methodology for probe sweeps, without which this study would have reported a false positive (Sec. 5). (3) The finding: attribution is not linearly decodable (Q1, Q2 negative) while a working positive control proves the label recoverable and the plan present (Sec. 4); the policy nevertheless reacts, but to visual novelty rather than to command–outcome mismatch (Q3; Sec. 4.2). (4) A pre-registered baseline for post-training comparisons: the same pipeline runs unchanged on RL-fine-tuned siblings of this policy (Sec. 7).

2 Related work

Efference copies and corollary discharge. The reafference principle [von Holst and Mittelstaedt, 1950] and corollary discharge [Sperry, 1950] describe circuitry that predicts the sensory consequences of self-generated action; Crapse and Sommer [2008] review its ubiquity across taxa. We borrow the concept as a representational hypothesis about artificial policies.

Do policies internalize actions? Li et al. [2026] document *action-blindness* in latent world models: predictions for different actions become indistinguishable even as training loss improves. Ye [2026] show minimal predictive agents can compensate for action hijacking without encoding agency. Our study ports this intervention-based test to a 7B VLA. Complementarily, world-model-based post-training such as WMPO [Zhu et al., 2025] *assumes* faithful action-conditioning; our protocol provides the measurement side.

Probing methodology. Linear probes [Alain and Bengio, 2016] require careful null calibration [Hewitt and Liang, 2019, Belinkov, 2022]. We extend control tasks with a *selection-aware* floor for best-of- N cell sweeps, which proved decisive (Sec. 5).

Table 1: Probe ladder. $h(t)/h(t+1)$: hidden states before/after the manipulated transition; a : commanded chunk; Δs : simulator state change.

Probe	Input	Question
P0	hidden state, shuffled labels	selection-aware chance floor
P1	$h(t+1)$	anomaly trace without efference (Q1)
P3	$a \oplus h(t+1)$	efference-augmented attribution (Q2)
P4	$h(t) \oplus h(t+1)$	internal-only efference availability (Q2)
P2	$a \oplus \Delta s$ (concatenated)	linear mechanical baseline
P2r	comparison features of $(a, \Delta s)$	mechanical ceiling / positive control
$C_{\text{cmd}}/C_{\Delta s}/C_{\text{phase}}$	$a / \Delta s / \text{phase alone}$	leakage controls

3 Setup

Model and tasks. OpenVLA-OFT (7B), SFT on LIBERO-Goal, evaluated at its released sampling protocol (temperature 1.6) through a standard evaluation stack; no proprioception, no wrist camera, action chunks of $K=8$ steps $\times$ 7 DoF. Main run: all 10 tasks $\times$ 30 fixed initial states = 300 episodes, 8,748 policy calls, 980 hijacks; 20% of episodes are never hijacked. Total collection cost including pilots and control arms: 1.9 GPU-hours on one 24 GB workstation GPU.

Causal hijack. At each eligible call the executed chunk is either the policy’s own (SELF) or a transform of it (HIJACK), i.i.d. with $p=0.25$ after a 4-call warm-up, never on consecutive calls, so the state one call later reflects exactly one manipulated transition (Fig. 1). SWAP, the primary transform, executes another same-task episode’s commanded chunk: replacement actions retain natural task statistics, so a probe cannot win by detecting implausible kinematics. MIRROR and FREEZE are collected as smaller control arms (50 episodes each) on the five tasks where the intervention perturbs mechanics most, and the SWAP data are re-analyzed restricted to those same tasks (task-matched arm).

Verified capture. Hidden states are recorded at layers $\{0, 4, \dots, 32\}$ under three pools: mean over context tokens, last pre-action token, and mean over the 56 positions from which the model decodes actions. A runtime self-check reproduces the model’s own action log-probabilities from the captured slice (max deviation 4.8×10^{-7}), guarding against indexing errors.

Probes and the selection-aware floor. Each probe sample is a pair of consecutive calls: positives follow a hijack; negatives follow a SELF transition *from the same episode*, phase-matched by minimum-cost assignment on call index (1,716 balanced samples; 858 pairs). Probes are ℓ_2 -regularized logistic regressions with 5-fold CV grouped by episode and inner-split regularization selection. Table 1 defines the ladder. Because each probe is reported at its best of 27 layer $\times$ pool cells, comparing to 0.5 overstates it: our floor is the maximum over the *same* 27 cells of a shuffled-label probe (P0), complemented by a fixed-cell permutation null. In the main run the floor is 0.536; effects below ≈ 0.04 balanced accuracy above chance are undetectable at this n .

Auxiliary measurements. *E1 (plan readout)*: ridge regression of the commanded 56-dim chunk from $h(t)$ (5-fold CV R^2). *E3 (behavioral reaction)*: paired differences (hijack – matched SELF) in next-call action-token entropy and chosen-chunk log-probability, tested with episode-level Wilcoxon signed-rank. A *corrective-behavior* metric (does the next command re-assert motion blocked by FREEZE?) is reported only after replacing the next command with its AR(1) innovation, since the raw version is confounded by command autocorrelation (Sec. 5).

4 Results

4.1 Attribution is not linearly decodable—with a working positive control

In the main run (Fig. 2a), $P1 = 0.525 \pm 0.025$, $P3 = 0.525 \pm 0.019$, and $P4 = 0.533 \pm 0.012$, all at or below the selection-aware floor of 0.536 ± 0.011 ; leakage controls sit at chance. The null is not unrecoverability: P2r reaches 0.690 ± 0.044 (AUROC 0.77) in the same data, rising to 0.726 (task-matched SWAP), 0.805 (MIRROR), and 0.875 ± 0.008 (FREEZE, AUROC 0.93) as the transform breaks the command $\rightarrow$ outcome link more thoroughly (Fig. 2b)—while hidden-state probes track their own per-arm floors in every arm (Appendix A), and held-out-task transfer of P1/P3/P4 is flat

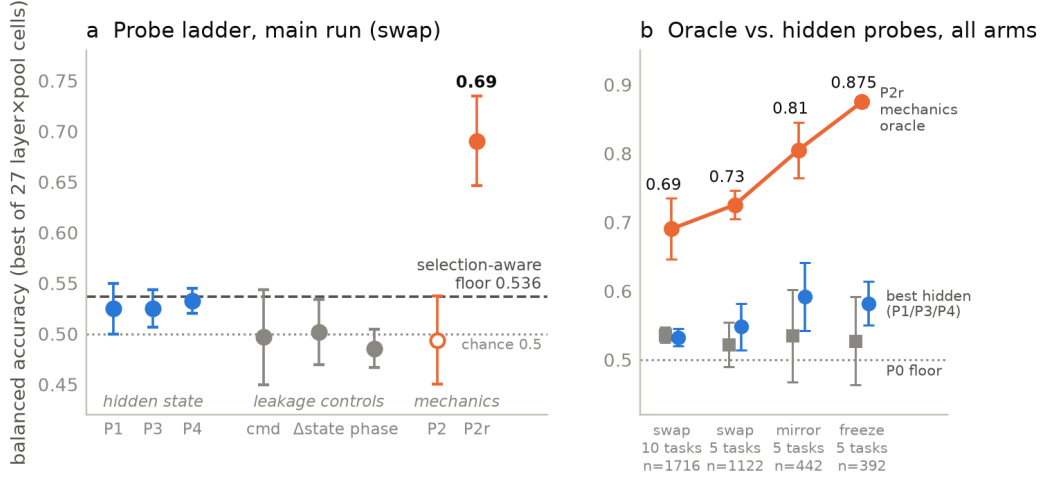


Figure 2: **(a)** Main-run ladder at each probe’s best of 27 cells ($\pm$ fold sd). Hidden-state probes (blue) sit at the selection-aware floor; leakage controls at chance; only the comparison-feature oracle P2r clears the floor (P2, open marker, is the concatenation baseline— agreement between two vectors is not a linear feature of their concatenation). **(b)** Across arms, P2r rises as the transform degrades the command→outcome link, while the best hidden probe tracks each arm’s own floor.

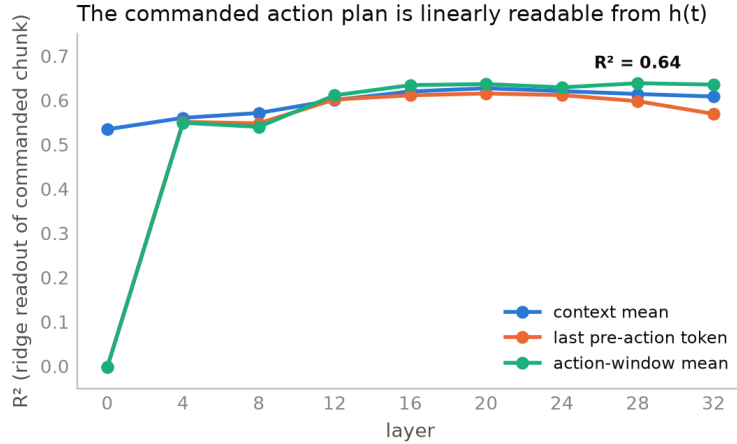


Figure 3: E1: the commanded 56-dim chunk is linearly readable from the pre-action state (plateau R^2 0.62–0.64 at layers 16–28; the two layer-0 pools at $R^2 \approx 0$ are structurally degenerate). The motor plan is *present*; it is not *usable* for attribution.

at 0.50. Under the pre-registered hypothesis ledger the sharpest cell is P4: handing the probe the model’s own “before” and “after” states does not help.

P4’s null is informative only if $h(t)$ actually contains the plan. It does (Fig. 3): ridge readout of the commanded chunk reaches $R^2=0.64$, with regularization interior to the search grid in 25/27 cells. The ingredients for a self-vs.-world comparison are linearly present; the comparison is not.

4.2 The policy reacts—to novelty, not to its own agency

After a SWAP, the next call’s action distribution shifts: entropy $+0.010$ nats, chosen-chunk log-probability -1.12 (episode-level Wilcoxon $p=2.4 \times 10^{-3}$ and 6.0×10^{-4} ; $n=217$ episodes, 858 pairs), reproducing in the task-matched arm ($p=4.5 \times 10^{-3}$, 3.1×10^{-3}). The fingerprint across arms is the interesting part (Fig. 4): if the policy compared outcomes to commands, surprise should

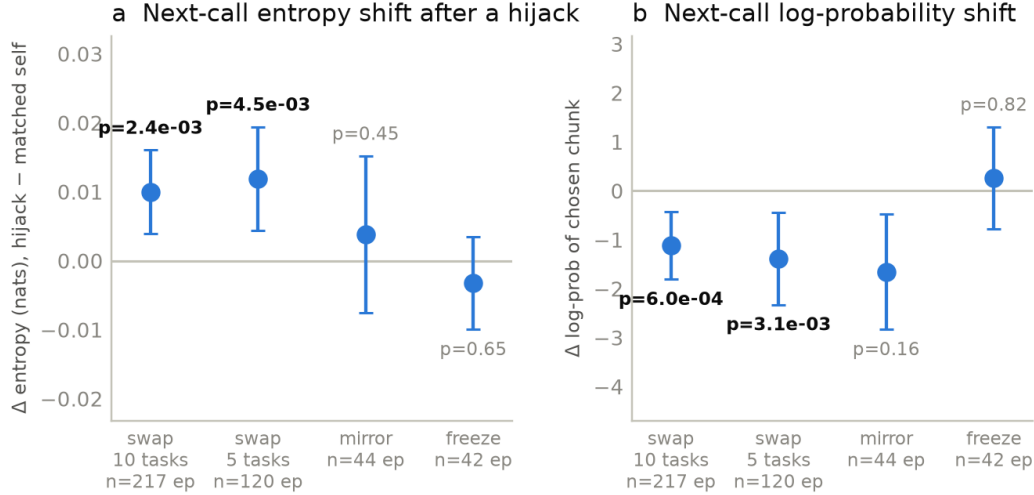


Figure 4: E3 reaction on the call after a hijack: paired difference in next-call entropy (a) and chosen-chunk log-probability (b). Points: pair-level means; error bars: ± 1 SE across episodes (the Wilcoxon unit); bold p = significant. SWAP shifts on both metrics in both analyses; FREEZE—the most mechanically decodable mismatch—does not.

peak for FREEZE, whose mismatch is most decodable (P2r 0.875). Instead FREEZE is flat with both point estimates on the wrong side (entropy -0.003 , $p=0.65$; log-prob $+0.27$, $p=0.82$): a frozen arm leaves a perfectly familiar—just stationary—scene. MIRROR, which does create visually novel (reversed) motion, trends like SWAP but is underpowered ($n=44$ episodes; entropy $+0.004$, $p=0.45$; log-prob -1.66 , $p=0.16$). The reaction tracks how visually unusual the outcome is, not how far execution deviated from the command.

Two supporting observations. Episode success barely moves under hijacking (0.411 vs. 0.458 clean, n.s.): the closed loop absorbs $p=0.25$ hijacks. And after FREEZE, the next command re-asserts the blocked motion beyond what command autocorrelation predicts (innovation-corrected undo-alignment 0.47 vs. 0.22 SELF baseline)—exactly what visual feedback control predicts, with no efferece required.

Together: the representation exposes the plan ($R^2=0.64$) and the mechanics expose the mismatch (P2r up to 0.875), yet no linear function of the representation—even given the plan—recovers “that wasn’t me,” while behavior shifts with outcome novelty. This is the substrate-level version of Ye’s compensation-without-encoding, in a 7B post-trained policy.

5 Four ways this experiment almost returned the wrong answer

Best-cell selection inflates ladders. P1’s best cell (0.525) beats the fixed-cell permutation null ($p_{95}=0.516$)—naively “weakly above chance.” But given the same 27 tries, shuffled-label P0 reaches 0.536. At pilot scale ($n=138$) the selection gap was 0.06–0.12: small- n probe sweeps without a selection-matched null overclaim by construction.

A linear “oracle” cannot compute agreement. Logistic regression on $a \oplus \Delta s$ is bilinear-blind; on synthetic data with fully determined labels it scores 0.52 while explicit comparison features score 0.94. Real data behaved as predicted per transform (P2 = 0.494 for mean-preserving SWAP vs. 0.570/0.538 for MIRROR/FREEZE, which leak through the marginal). P2r is the honest ceiling; a naive acceptance gate on P2 would have misfired in both directions.

Command autocorrelation fakes “correction.” The intuitive undo-alignment metric reduces to $+\langle a_{\text{prev}}, a_{\text{next}} \rangle$ under FREEZE, so any smooth policy scores as corrective: a provably open-loop simulation with $\rho=0.9$ scores $+0.83$ raw and 0.00 after AR(1) innovation correction. On real data ($\rho=0.63$ –0.72) the raw metric inflates 2–4 $\times$.

Task competence gates intervention strength. On a task the policy cannot do it barely moves and emits near-identical chunks across episodes, making SWAP a near-no-op (first pilot: P2r 0.585—the oracle itself near chance). Three single-variable pilots isolated this; competent tasks restore P2r to 0.749 with controls at chance. Weak-intervention tasks dilute rather than bias, but undiagnosed they would have made the main run uninterpretable.

6 Limitations

Linear probes only: nonlinear or distributed attribution codes are untouched, and the cheapest falsification is an MLP probe on the same frozen features. Power: effects below ≈ 0.04 balanced accuracy are invisible, and E3 proves a small distributional signal exists, so a weak per-instance code inside that window survives. Small control arms (~ 43 episodes each): hidden-probe excursions to ≈ 0.58 stay within those arms’ noisy floors and shrink in the larger SWAP arms, and MIRROR’s same-direction E3 trend is not independently significant—but both warrant scaled replication. Phase matching is imperfect (negatives sit a median of 2 calls earlier; the bias is anti-conservative for positives, and $C_{\text{phase}} = 0.486$ shows phase is not linearly exploitable, so the null stands *a fortiori*). One model, one suite, one checkpoint; rollouts capped at 320 steps.

7 Discussion: a baseline for post-training comparisons

For this SFT checkpoint, compensation is visual servoing; no efference-copy substrate is linearly readable even though the plan is. The pipeline was built to make the next comparisons cheap: (i) the same battery on RL fine-tuned siblings—on-policy methods such as world-model-based GRPO [Zhu et al., 2025, Shao et al., 2024] train precisely on the consequences of self-generated actions, and whether they build the monitoring substrate SFT lacks is a pre-registered open question; (ii) the same intervention grammar aimed at action-conditioned world models themselves, where action-blindness [Li et al., 2026] would silently corrupt imagined on-policy training; (iii) nonlinear probes and activation patching at the action-decoding window. Code, data, and the append-only preregistration log will be released with the camera-ready version.

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A Full ladders per arm

Table 2: Balanced accuracy at each probe’s best layer $\times$ pool cell ($\pm$ fold sd); floors and permutation nulls computed per arm. E3 rows: paired mean difference (episode-level Wilcoxon p).

	SWAP 10 tasks ($n=1716$)	SWAP 5 tasks ($n=1122$)	MIRROR ($n=442$)	FREEZE ($n=392$)
P0 floor	0.536 ± 0.011	0.522 ± 0.032	0.535 ± 0.067	0.528 ± 0.064
perm. null p_{95} (fixed cell)	0.516	0.518	0.535	0.548
P1	0.525 ± 0.025	0.540 ± 0.033	0.583 ± 0.058	0.582 ± 0.032
P3	0.525 ± 0.019	0.548 ± 0.034	0.586 ± 0.017	0.579 ± 0.045
P4	0.533 ± 0.012	0.523 ± 0.016	0.592 ± 0.049	0.552 ± 0.068
P2 (concat)	0.494 ± 0.044	0.491 ± 0.028	0.570 ± 0.016	0.538 ± 0.032
P2r (comparison)	0.690 ± 0.044	0.726 ± 0.021	0.805 ± 0.040	0.875 ± 0.008
C_{cmd}	0.497	0.487	0.534	0.505
$C_{\Delta s}$	0.502	0.474	0.599	0.567
C_{phase}	0.486	0.522	0.529	0.510
E3 entropy $\Delta(p, n_{\text{ep}})$	+0.010 (2.4e-3, 217)	+0.012 (4.5e-3, 120)	+0.004 (0.45, 44)	−0.003 (0.65, 42)
E3 log-prob $\Delta(p)$	−1.12 (6.0e-4)	−1.38 (3.1e-3)	−1.66 (0.16)	+0.27 (0.82)
success rate	0.420	0.253	0.180	0.360

Cross-task transfer, main run (train on ~ 7 tasks, test held-out): P1 0.505, P3 0.504, P4 0.501. E1 readout: R^2 0.535 at layer 0 (context mean) rising to a 0.62–0.64 plateau at layers 16–28 (best 0.639); regularization interior to the grid in 25/27 cells.

B Qualitative examples (random, not cherry-picked)

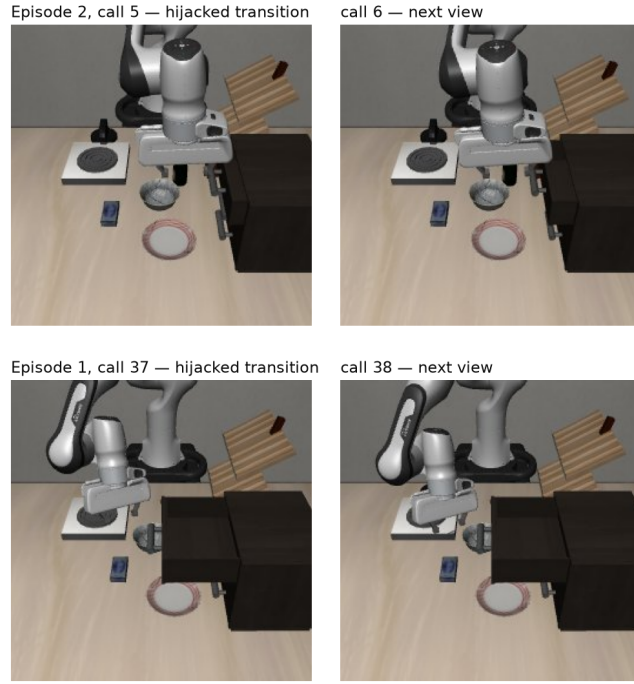


Figure 5: Two hijacked transitions drawn with a pre-registered seed from the 24 hijacks with saved frames in the task-3 pilot. **Top** (episode 2, call 5): commanded net $\Delta y + 5.7$, $\Delta z - 4.4$, gripper channel $+1$; executed $\Delta y + 2.5$, $\Delta z - 6.2$, gripper -1 . The next call became *more* confident (entropy 0.091 vs. episode self-mean 0.240; log-prob -6.5 vs. -20.8); the episode succeeds. **Bottom** (episode 1, call 37): commanded up ($\Delta z + 2.4$), executed down (-5.0); entropy 0.409 vs. 0.283, log-prob -32.0 vs. -26.7 ; the episode fails. One of two random draws goes against the population trend: within-episode scatter (entropy sd ≈ 0.10) dwarfs the $+0.010$ mean shift, which is why a real distributional effect coexists with floor-level per-instance decodability.